

BUSINESS REQUIREMENTS DOCUMENT

E-Commerce Sales & Customer Behaviour Analysis

Revenue Loss Identification & Customer Retention Strategy

Field	Details
Document Author	Kapil Waghumbare
Role	Business Analyst (Final Year, BE Computer Engineering)
Tools Used	MySQL Power BI Microsoft Excel DAX
Domain	E-Commerce / Retail
Dataset	15,104 Orders 2,430 Customers Jan–Dec 2026
Version	v1
Date	March 2026
Status	FINAL

1. Executive Summary

This Business Requirements Document presents a comprehensive analysis of an e-commerce retail platform covering 15,104 transactions from 2,430 unique customers across January to December 2026. The analysis was performed using MySQL for data querying, Power BI for dashboard visualization, and Excel for reporting.

Key findings:

- Total platform revenue: ₹12,78,99,261 (delivered orders only)
- Revenue lost to cancellations: ₹4,91,43,990 (22.1% cancellation rate)
- Revenue lost to returns: ₹4,19,73,244 (20.0% return rate)
- Total combined revenue loss: ₹9,11,17,234 — 41.6% of potential total revenue
- Electronics is the highest revenue category at ₹9.17Cr but also has high return rate (19.3%)
- COD payment method has the highest cancellation rate among all payment methods
- Average Order Value: ₹14,635 for delivered orders

2. Business Context & Problem Statement

2.1 Background

The client is a mid-size Indian e-commerce retailer operating across 15 major cities, offering products across 6 categories: Electronics, Fashion, Home & Kitchen, Beauty, Sports, and Books. The business operates a multi-channel model accepting 5 payment methods and serving customers across Mobile, Desktop, and Tablet devices.

2.2 Problem Statement

Despite healthy order volumes of 15,104 orders in 2026, the business is losing 41.6% of potential revenue through cancellations and returns. The management requires data-backed analysis to:

- Quantify the total financial impact of cancellations and returns
- Identify which categories, cities and payment methods drive highest loss
- Understand customer behaviour patterns that predict cancellation/return likelihood
- Develop actionable recommendations to recover lost revenue

2.3 Business Objectives

Objective	Metric	Target
Reduce cancellation rate	Cancellation % of total orders	From 22.1% to below 12%
Reduce return rate	Return % of total orders	From 20.0% to below 12%
Increase monthly revenue	Monthly delivered revenue	+15% in 6 months
Improve customer LTV	Avg spend per customer	+20% for Platinum segment
Shift COD to prepaid	% prepaid vs COD orders	COD below 15%

3. Project Scope

3.1 In Scope

- Analysis of 15,104 e-commerce orders (Jan–Dec 2026)
- Analysis of 2,430 unique customers with segmentation
- Revenue performance by category, city, payment method and device
- Cancellation and return pattern analysis
- Customer segmentation — Platinum, Gold, Silver, Bronze
- Monthly revenue trend and seasonal analysis
- 7 business recommendations with expected financial impact

3.2 Out of Scope

- Real-time data integration or live dashboard deployment
- Supplier or inventory management analysis
- Social media or marketing channel attribution

4. Data Overview

4.1 Orders Dataset — 15,104 Records

Field	Type	Description
order_id	VARCHAR(10)	Unique order identifier (ORD10001 – ORD25104)
customer_id	VARCHAR(10)	Customer identifier — links to Customers table
order_date	DATE	Order placement date (Jan–Dec 2026)
product_name	VARCHAR(50)	Name of purchased product (48 unique products)
category	VARCHAR(30)	Product category (6 categories)
quantity	INT	Number of units ordered (1–5)
unit_price	DECIMAL(10,2)	Price per unit in INR
discount_pct	INT	Discount applied (0–20%)
final_amount	DECIMAL(10,2)	Final order value after discount
payment_method	VARCHAR(20)	Payment type (5 methods)
device_type	VARCHAR(10)	Device used (Mobile/Desktop/Tablet)
city	VARCHAR(20)	Customer city (15 cities)
order_status	VARCHAR(15)	Delivered / Cancelled / Returned

4.2 Customers Dataset — 2,430 Records

Field	Type	Description
customer_id	VARCHAR(10)	Primary Key — links to Orders table
customer_name	VARCHAR(50)	Full name of customer
gender	VARCHAR(10)	Male / Female
age	INT	Customer age (18–58)
email	VARCHAR(100)	Customer email address
phone	VARCHAR(15)	Customer phone number
city	VARCHAR(20)	Customer home city
join_date	DATE	Date customer joined platform
total_orders	INT	Total orders placed by customer
total_spent	DECIMAL(12,2)	Total amount spent (delivered orders only)
first_order_date	DATE	Date of first order
last_order_date	DATE	Date of most recent order
avg_rating	DECIMAL(3,1)	Average rating given by customer
segment	VARCHAR(15)	Platinum / Gold / Silver / Bronze

5. Key Findings & SQL Insights

5.1 Business Health Overview (Query 1)

Metric	Value	Insight
Total Orders	15,104	Strong order volume across 12 months
Delivered Orders	8,739 (57.9%)	Less than 60% delivery — critical issue
Cancelled Orders	3,340 (22.1%)	2x industry benchmark of ~10%
Returned Orders	3,025 (20.0%)	2x industry benchmark of ~10%
Total Revenue Earned	₹12,78,99,261	Realized revenue
Cancelled Loss	₹4,91,43,990	Major revenue leakage
Returned Loss	₹4,19,73,244	Major revenue leakage
Total Revenue Loss	₹9,11,17,234	41.6% of potential revenue lost
Avg Order Value	₹14,635	Healthy AOV — Electronics driving this

5.2 Category Performance (Query 2)

Category	Revenue (₹)	Orders	Cancel Rate	Return Rate
Electronics	9,16,57,382	2,529	22.1%	19.3%
Home & Kitchen	1,41,37,628	2,473	23.0%	20.1%
Fashion	1,03,76,054	2,438	21.8%	19.6%
Sports	68,77,130	2,543	22.2%	19.8%
Beauty	27,97,971	2,558	21.7%	20.3%
Books	20,53,096	2,563	21.9%	21.0%

5.3 Customer Segmentation (Query 6)

Segment	Criteria	Count	Action
Platinum	> ₹50,000 spent	674	VIP loyalty, priority delivery, exclusive deals
Gold	₹20,000–50,000	585	Targeted upsell, early access campaigns
Silver	₹8,000–20,000	587	Re-engagement emails, loyalty points
Bronze	< ₹8,000	584	First purchase incentive, onboarding

6. Power BI Dashboard Specifications

Visual	Type	Metrics
Revenue KPI Cards (4)	Card	Total Revenue Cancelled Loss Returned Loss Total Loss
Monthly Revenue Trend	Line Chart	order_date (Month) vs Total Revenue DAX measure
Revenue by Category	Bar Chart	category vs Total Revenue — sorted descending
Order Status Distribution	Donut Chart	order_status count — Delivered/Cancelled/Returned
Revenue by City	Bar Chart	city vs Total Revenue — top 15 cities
Cancellation by Payment	Bar Chart	payment_method vs Cancellation Rate DAX measure

6.1 DAX Measures Used

```
Total Revenue = CALCULATE(SUM(orders[final_amount]), orders[order_status] = "Delivered")
Cancelled Loss = CALCULATE(SUM(orders[final_amount]), orders[order_status] = "Cancelled")
Returned Loss = CALCULATE(SUM(orders[final_amount]), orders[order_status] = "Returned")
Total Loss = [Cancelled Loss] + [Returned Loss]
Cancellation Rate = DIVIDE(CALCULATE(COUNT(orders[order_id]), orders[order_status]="Cancelled"), COUNT(orders[order_id]))
```

7. Business Recommendations

#	Finding	Recommendation	Expected Impact	Priority
1	Cancellation 22.1% — ₹4.91Cr lost	Mandatory confirmation + live ETA before order placement	Save ₹1.5–2Cr/year	HIGH
2	COD highest cancel rate	₹29 COD fee OR 5% prepaid discount	Shift 20% COD to prepaid	HIGH
3	Return rate 20% — ₹4.19Cr lost	Product videos, 360 images, size guides	Reduce returns by 20%	HIGH
4	Electronics high returns (19.3%)	Product comparison tool + spec checklist	Save ₹50–70L	HIGH
5	674 Platinum customers	Platinum loyalty — priority delivery + early access	LTV +25–30%	HIGH
6	Oct–Nov seasonal spike	Pre-stock Sep; loyalty early access Oct 1	+15–20% Q4 revenue	MEDIUM
7	Tier-2 cities underperforming	City-specific promotions for Surat, Jaipur, Coimbatore	Tier-2 revenue +20%	LOW

8. Success Metrics

Metric	Current (2026)	6-Month Target	Tool
Cancellation Rate	22.1%	< 12%	MySQL + Power BI
Return Rate	20.0%	< 12%	MySQL + Power BI
Monthly Revenue	₹1.07Cr avg	₹1.23Cr (+15%)	Monthly SQL report
COD Share	~22%	< 15%	Payment Analysis Query
Platinum Customers	674	800+	Customer Segmentation
Avg Order Value	₹14,635	₹16,000+	Business Health Query

9. Appendix — SQL Queries

Query	Title	Business Question
Query 1	Business Health Overview	Total revenue vs cancellation/return losses
Query 2	Category Performance	Which categories earn most and lose most?
Query 3	Monthly Revenue Trend	Seasonal patterns in revenue?
Query 4	Payment Method Analysis	Which payment method has highest cancel rate?
Query 5	City Revenue Analysis	Which cities generate most revenue?
Query 6	Customer Segmentation	Who are top 20 most valuable customers?

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